

LAB # 10

JUnit Testing

Lab Task:

- Add another test case for countA function (which is given in code).

```
public class Main {  
  
    // square method  
    public static int square(int x) {  
        return x * x;  
    }  
  
    // countA method  
    public static int countA(String word) {  
        int count = 0;  
        for (int i = 0; i < word.length(); i++) {  
            if (word.charAt(i) == 'a' || word.charAt(i) == 'A') {  
                count++;  
            }  
        }  
        return count;  
    }  
  
    // main method for testing  
    public static void main(String[] args) {  
  
        // Test square()  
        int squareResult = square(5);  
        System.out.println("Square Test:");  
        System.out.println("Expected: 25");  
        System.out.println("Actual : " + squareResult);  
  
        // Test countA()  
        int countAResult = countA("AnApple");  
        System.out.println("\nCountA Test:");  
        System.out.println("Expected: 3");  
        System.out.println("Actual : " + countAResult);  
    }  
}
```

```
Square Test:  
Expected: 25  
Actual : 25
```

```
CountA Test:  
Expected: 3  
Actual : 2
```

- Make new project, make a class. Add 2 methods in it. One method will find the max integer present in the input integer array. The other method will find the min integer. Now create test cases for both these methods and test your code. Follow all the steps as mentioned above in the manual.

```
public class Main {  
    // Method to find maximum value in array  
    public static int findMax(int[] arr) {  
        int max = arr[0];  
        for (int i = 1; i < arr.length; i++) {  
            if (arr[i] > max) {  
                max = arr[i];  
            }  
        }  
        return max;  
    }  
  
    // Method to find minimum value in array  
    public static int findMin(int[] arr) {  
        int min = arr[0];  
        for (int i = 1; i < arr.length; i++) {  
            if (arr[i] < min) {  
                min = arr[i];  
            }  
        }  
        return min;  
    }  
  
    // Testing both methods  
    public static void main(String[] args) {  
        int[] numbers = {10, 5, 20, 3, 15};  
  
        // Test findMax  
        int maxResult = findMax(numbers);  
        System.out.println("Testing findMax()");  
        System.out.println("Expected Max: 20");  
        System.out.println("Actual Max : " + maxResult);  
  
        return max;  
    }  
  
    // Method to find minimum value in array  
    public static int findMin(int[] arr) {  
        int min = arr[0];  
        for (int i = 1; i < arr.length; i++) {  
            if (arr[i] < min) {  
                min = arr[i];  
            }  
        }  
        return min;  
    }  
  
    // Testing both methods  
    public static void main(String[] args) {  
        int[] numbers = {10, 5, 20, 3, 15};  
  
        // Test findMax  
        int maxResult = findMax(numbers);  
        System.out.println("Testing findMax()");  
        System.out.println("Expected Max: 20");  
        System.out.println("Actual Max : " + maxResult);  
  
        // Test findMin  
        int minResult = findMin(numbers);  
        System.out.println("\nTesting findMin()");  
        System.out.println("Expected Min: 3");  
        System.out.println("Actual Min : " + minResult);  
    }  
}
```

```
Testing findMax()  
Expected Max: 20  
Actual Max : 20  
  
Testing findMin()  
Expected Min: 3  
Actual Min : 3
```

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 LAB#9 SCD SARA.docx	Add files via upload	13 minutes ago
 Lab#5 scd.pdf	Add files via upload	last month
 Lab#6 scd sara.pdf	Add files via upload	last month
 lab#1 scd sara.pdf	Add files via upload	last month
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